

linear modeling

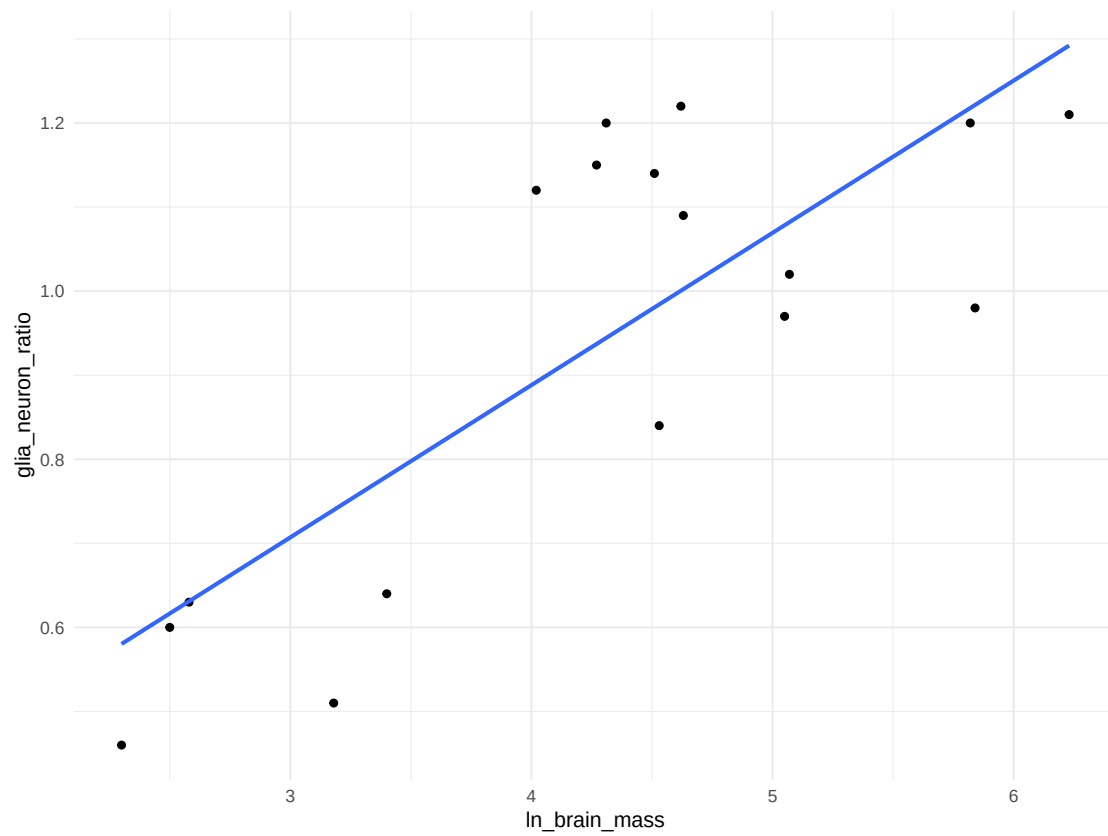
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Problem 3

```
##  
## Call:  
## lm(formula = glia_neuron_ratio ~ ln_brain_mass, data = .)  
##  
## Coefficients:  
## (Intercept) ln_brain_mass  
## 0.1637 0.1811
```

Problem 3a



```
##
```

```
## Call:
## lm(formula = glia_neuron_ratio ~ ln_brain_mass, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.24150 -0.12030 -0.01787  0.15940  0.25563
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.16370    0.15987   1.024 0.322093
## ln_brain_mass 0.18113    0.03604   5.026 0.000151 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1699 on 15 degrees of freedom
## Multiple R-squared:  0.6274, Adjusted R-squared:  0.6025
## F-statistic: 25.26 on 1 and 15 DF,  p-value: 0.0001507

##              1
## 1.471458
```

Problem 3b

The predicted glia-neuron ratio for humans given the non human data is 1.47.

Problem 3c

The interval for the mean glia-neuron ratio at the given brain mass is preferable because it takes into account the data we have observed.

Problem 3d

$$\hat{Y}_h = b_0 + b_1 * X_h$$

$$s^2(\hat{Y}_h) = MSE \left[\frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2} \right]$$

$$s^2(\hat{Y}_h) = 0.0774 * \left[\frac{1}{17} + \frac{(7.22 - 4.29)^2}{22.21} \right]$$

$$= .0774 * 0.4453566$$

$$= 0.0344706$$

$$s = \sqrt{s^2} = \sqrt{.034} = 0.1843909$$

95% CI:

$$\hat{Y}_h \pm t_{1-\alpha/2, n-2} * s(\hat{Y}_H)$$

$$= 1.47 \pm t_{.975, 15} * 0.18$$

$$= 1.47 \pm 2.1314495 * 0.18$$

= (1.0863391,1.8536609)

Since the human glia neuron ratio is within the 95% confidence interval of the predicted response there is no evidence that the human brain has an excessive glia-neuron ratio compared with other primates.

```
##          fit          lwr          upr
## 1 1.471458 1.036047 1.906869
```

Problem 3e

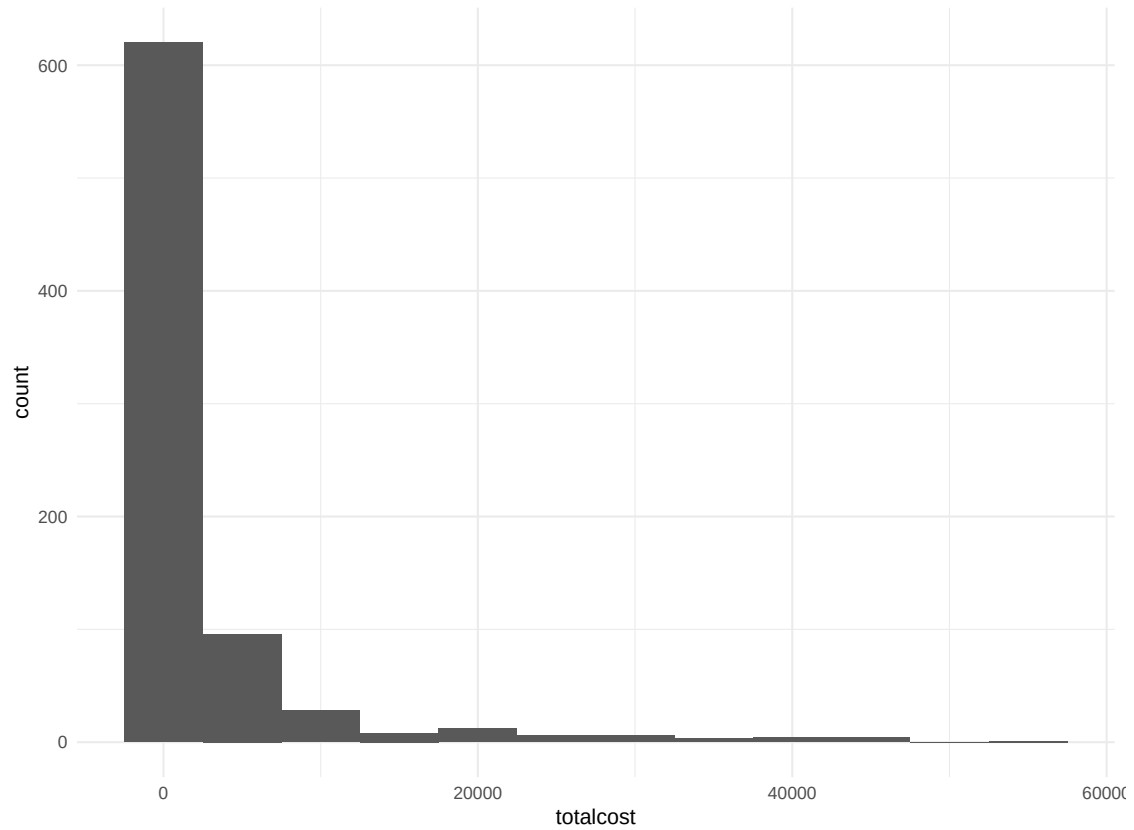
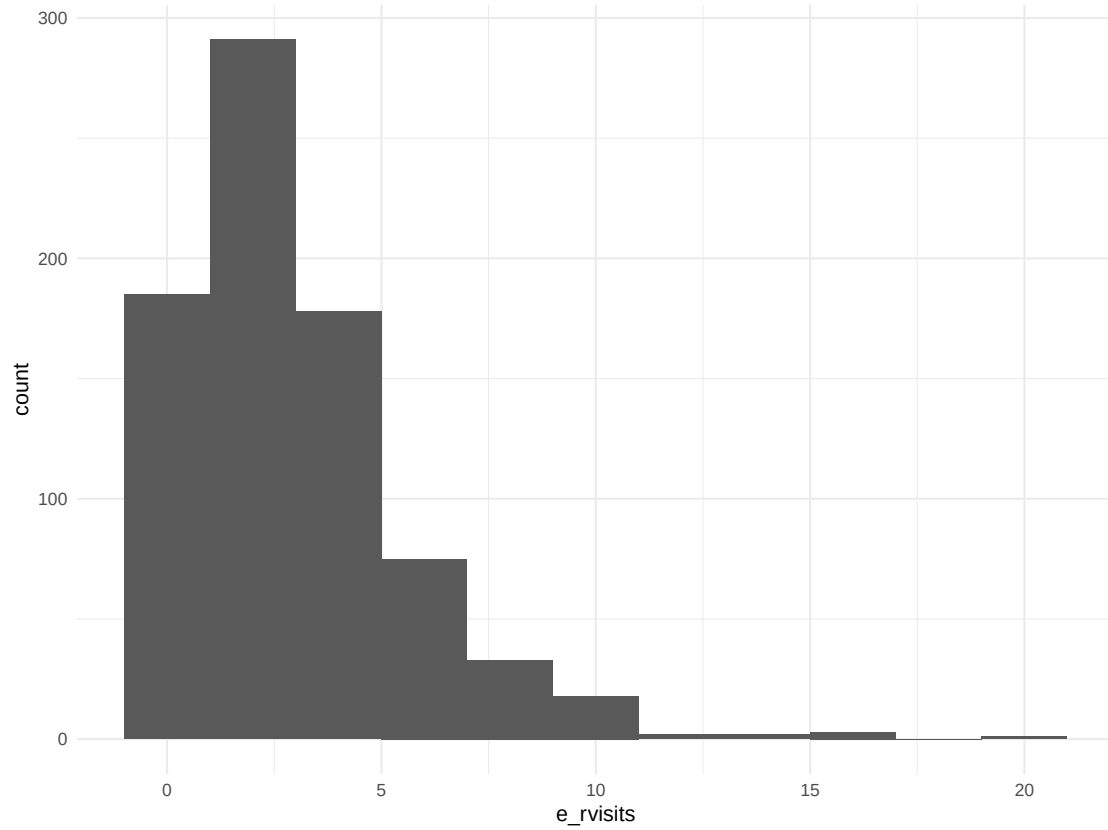
This is extrapolation since the prediction is outside the observed range. Extrapolation can provide accurate results when the prediction is close to the observed data. However, it may not be a good idea if the prediction is far from the observed data.

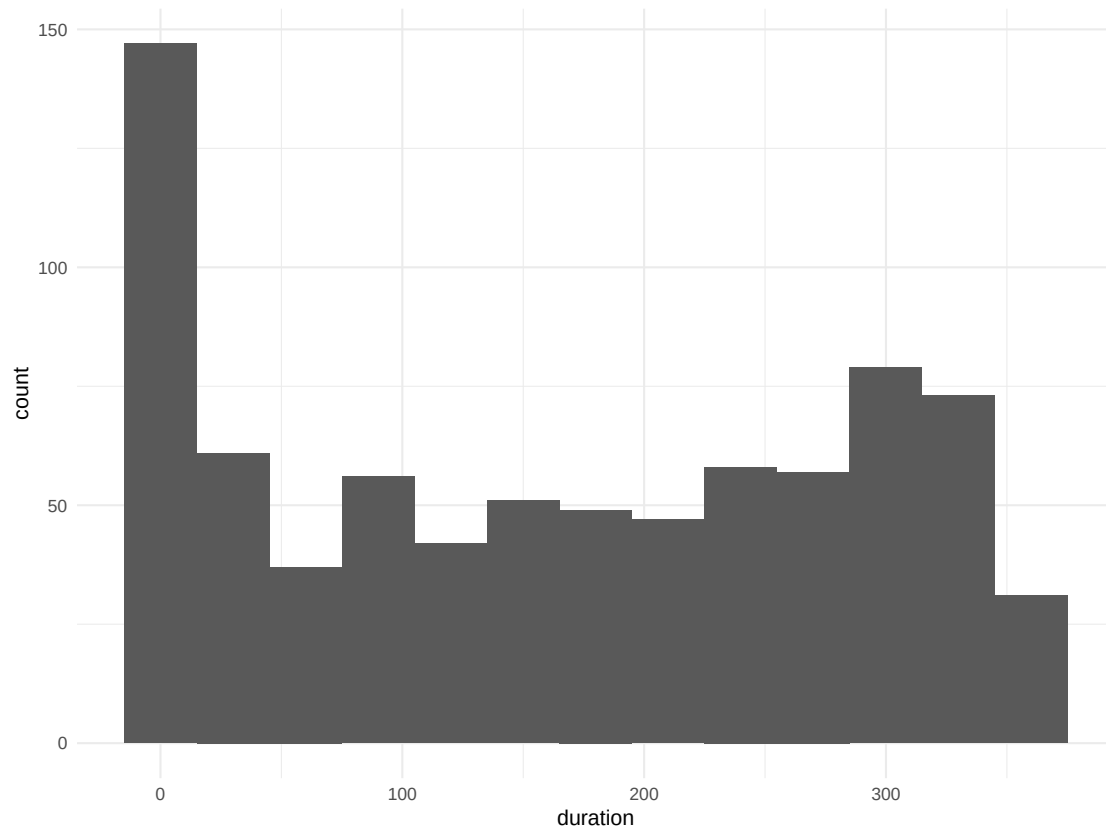
Problem 4

##Problem 4a

```
## Rows: 788 Columns: 10
## -- Column specification -----
## Delimiter: ","
## dbl (10): id, totalcost, age, gender, interventions, drugs, ERvisits, compli...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

This dataset is 788 rows long and 10 columns across. It records observations about patients diagnosed with heart disease. The main variables are total cost and number of emergency room visits. Other variables of interest are age, gender, complications, and duration of treatment. Other variables includes the number of drugs prescribed, and the number of comorbidities diagnosed.

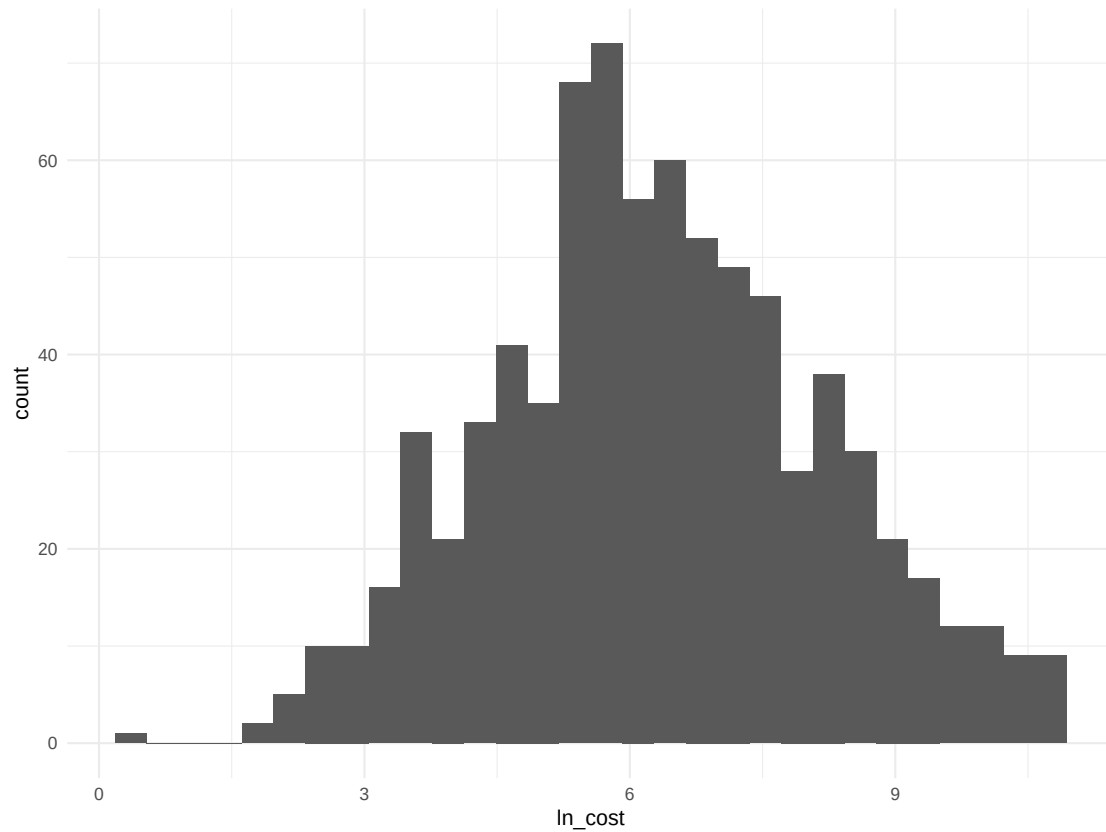




Problem 4b

The values of total cost are not normally distributed. One applies a log transformation.

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



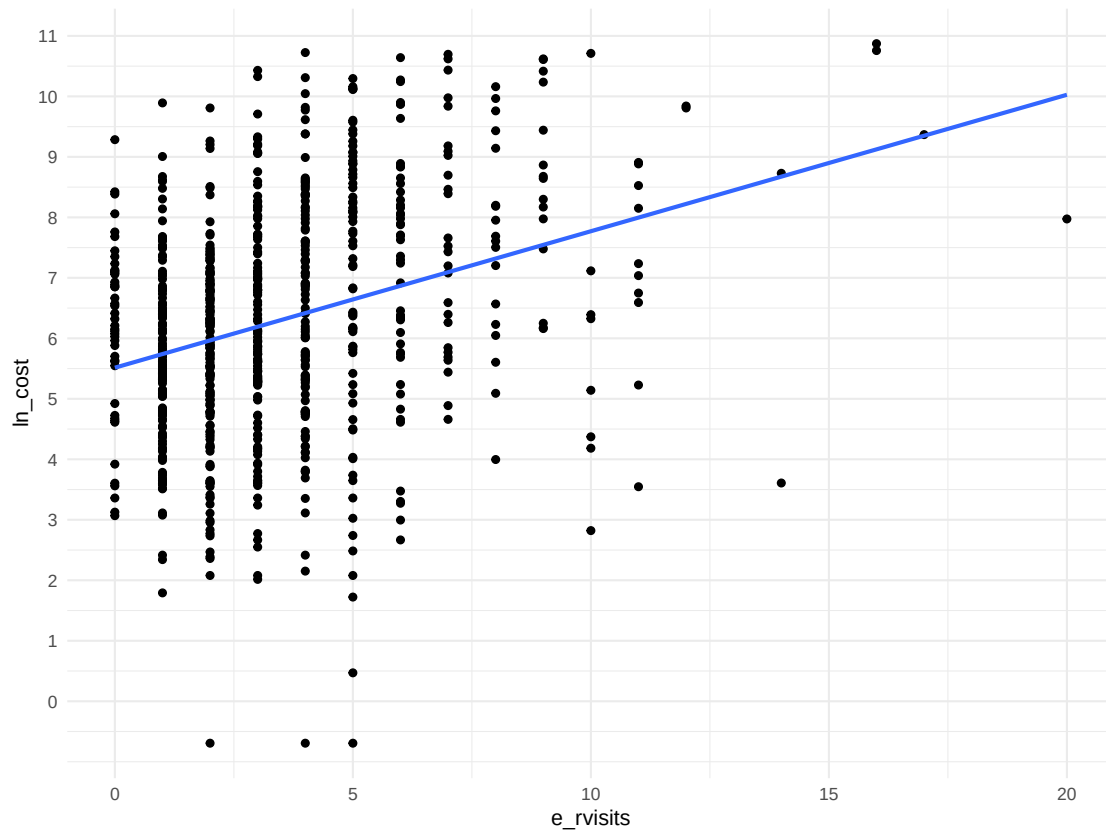
```
## # A tibble: 788 x 11
##       id totalcost  age gender interventions drugs e_rvisits complications
##   <dbl>   <dbl> <dbl> <dbl>         <dbl> <dbl>     <dbl>         <dbl>
## 1     1     179.   63     0           2     1         4           0
## 2     2     319.   59     0           2     0         6           0
## 3     3    9311.   62     0          17     0         2           0
## 4     4     281.   60     1           9     0         7           0
## 5     5   18727.   55     0           5     2         7           0
## 6     6     453.   66     0           1     0         3           0
## 7     7     323.   64     1           2     0         3           0
## 8     8    3874.   45     1           3     0         5           0
## 9     9    3244.   68     0           6     2         5           0
## 10    10     226.   64     1           3     0         2           0
## # i 778 more rows
## # i 3 more variables: comorbidities <dbl>, duration <dbl>, comp_bin <dbl>
```

Problem 4c

Add variables for the log of total cost and comp_bin.

Problem 4d

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
##
## Call:
## lm(formula = ln_cost ~ e_rvisits, data = .)
##
## Coefficients:
## (Intercept)    e_rvisits
##      5.5147      0.2256
```

The natural log of the cost increases by one unit for every .2256 rise in the number of visits.

To interpret this, one may back transform the slope by raising the natural number, e , to the power of the slope. So, for each emergency room visit, there is a $e^{.2256}$ times rise in cost. In other words, there is an $e^{.2256} - 1$ percent rise in cost for each emergency room visit. This is equivalent to 25.3074335 %.

```
##
## Call:
## lm(formula = ln_cost ~ e_rvisits, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.3358 -1.1168  0.0395  1.2899  4.3073
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  5.51469    0.10642  51.818   <2e-16 ***
## e_rvisits    0.22559    0.02462   9.161   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.822 on 786 degrees of freedom
## Multiple R-squared:  0.09648,    Adjusted R-squared:  0.09533
## F-statistic: 83.93 on 1 and 786 DF,  p-value: < 2.2e-16
```

This summary table shows the adjusted r-squared value is less than 0.1, indicating that the number of emergency room visits alone is a poor predictor for the total cost for a patient.

Problem 4e

```
##
## Call:
## lm(formula = ln_cost ~ e_rvisits + comp_bin, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.2067 -1.0717  0.0017  1.1957  4.4139
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.49804    0.10411  52.809   < 2e-16 ***
## e_rvisits    0.20311    0.02436   8.336 3.42e-16 ***
## comp_bin     1.71655    0.28273   6.071 1.98e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.782 on 785 degrees of freedom
## Multiple R-squared:  0.137,    Adjusted R-squared:  0.1348
## F-statistic: 62.31 on 2 and 785 DF,  p-value: < 2.2e-16
```

After back transforming on a log-scale, this summary table indicates that there is an 1.225195 times increase in the total cost for a patient per emergency room, or 22.5194982 visit and a 5.5655733 times increase in the total cost (456.557331 %) if there is a complication during treatment.

Problem 4ei

```
##
## Call:
## lm(formula = ln_cost ~ e_rvisits * comp_bin, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.218 -1.077  0.013  1.210  4.410
##
## Coefficients:
```



```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.47630    0.10635  51.492 < 2e-16 ***
## e_rvisits         0.20964    0.02522   8.312 4.15e-16 ***
## comp_bin          2.20238    0.56160   3.922 9.57e-05 ***
## e_rvisits:comp_bin -0.09765    0.09754  -1.001  0.317
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.782 on 784 degrees of freedom
## Multiple R-squared:  0.1381, Adjusted R-squared:  0.1348
## F-statistic: 41.88 on 3 and 784 DF, p-value: < 2.2e-16
```

To determine whether complications are an effect modifier of the association of number of emergency room visits with total cost, one determines if there is a significant difference in the estimated slope: $(.20964 - .22559)/.22559 = -0.01595/.22559 = -0.0707035$. A cutoff of ten percent is used, so this is not an effect modifier.

Problem 4eii

To determine whether complications are a confounder of the relationship between number of emergency room visits and total cost, there must be an association between the confounder and the exposure, being number of emergency room visits.

```
##
## Call:
## lm(formula = e_rvisits ~ comp_bin, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.0930 -1.3289 -0.3289  0.6711 16.6711
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.32886    0.09557  34.833 < 2e-16 ***
## comp_bin       1.76416    0.40911   4.312 1.82e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.608 on 786 degrees of freedom
## Multiple R-squared:  0.02311, Adjusted R-squared:  0.02187
## F-statistic: 18.6 on 1 and 786 DF, p-value: 1.822e-05
```

This summary table shows that there is a 1.76 increase in the number of emergency room visits if a complication occurs. However, the adjusted r-squared value indicates that a complication is a poor predictor of the number of emergency room visits. So, complication is probably not a confounder. On the other hand, the model including complications indicates there is a significant difference in the estimated slope: $(.20311 - .22559)/.22559 = -0.02248/.22559 = -0.0996498$.

Using a cutoff of 10%, this number is quite close.

Problem 4eiii

Since the presence of a complication was close to the cutoff for being a confounder and it raises the value of the adjusted r-squared by 50 percent in our model, it should be included in the model.

Problem 4f

```
##
## Call:
## lm(formula = ln_cost ~ e_rvisits + comp_bin + age + gender +
##     duration, data = .)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.1297 -1.0336 -0.1036  0.9586  4.3480
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.9240310   0.5167617  11.464 < 2e-16 ***
## e_rvisits    0.1744731   0.0228547   7.634 6.64e-14 ***
## comp_bin     1.5126547   0.2617067   5.780 1.08e-08 ***
## age         -0.0207402   0.0087826  -2.362  0.0184 *
## gender       -0.2191600   0.1404272  -1.561  0.1190
## duration     0.0057983   0.0004949  11.715 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.644 on 782 degrees of freedom
## Multiple R-squared:  0.268, Adjusted R-squared:  0.2633
## F-statistic: 57.26 on 5 and 782 DF, p-value: < 2.2e-16

## Analysis of Variance Table
##
## Model 1: ln_cost ~ e_rvisits + comp_bin
## Model 2: ln_cost ~ e_rvisits + comp_bin + age + gender + duration
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      785 2492.1
## 2      782 2113.9  3    378.24 46.642 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Problem 4fi

The MLR method using the variables for the number of emergency room visits, the presence of a complication, age, gender, and duration of treatment raises the adjusted r-squared by a factor of 2.

While the number of emergency room visits, the presence of a complication and the duration of treatment are associated with increased cost, age and being female are associated with a decrease in cost.

Gender and age account for a relatively low proportion of the adjusted r-squared

in the log cost. While duration accounts for a significant portion of the adjusted r-squared, the effect is very low - duration of treatment increases the price predictably, but at a very low rate. The cost to a patient would increase about 0.6% $((e^{0.0057983} - 1) \times 100\% = 0.5815143\%)$ per day. On the first day of the stay, a one day increase in length of stay is associated with an average increase in cost of about two dollars and a quarter $(e^{\beta_0} \times e^{\beta_1 * x} - e^{\beta_0})$
 $= e^{5.924} \times e^{0.005798} - e^{5.924} = 2.1741943$.
On the one-hundredth day it is associated with an increase of $e^{5.924} \times e^{0.005798 * 100} - e^{5.924} \times e^{0.005798 * 99}$
 $= 3.8599729$ or about four dollars.

Problem 4fi

Even though there is evidence to suggest the complex model accounts for more of the variability in our model according to a partial F-test, there is reason to believe a simple model may be more useful in practice.

Since the model with the number of emergency room visits and complications accounts for nearly half of the adjusted r-squared with three fewer variables, it may be more informative to include only these two variables.